

Developer Guide

River Hawk Consulting, LLC · IMAX HCD Prototype OT

August 8, 2026

CONTENTS

Developer Guide — IMAX Chat-Viewer (Phase 3 Angular build)

 Architecture

 Three things in here that are easy to undo by accident

 Dev topology

 Tests

 Ground rules for new work

 Deployment

 Repo map

Developer Guide — IMAX Chat-Viewer (Phase 3 Angular build)

UNCLASSIFIED — Deliverable ④. The Angular SPA at `angular/` is the delivery codebase; the React app at `src/` is the Phase 2 reference implementation (living spec) and ships unchanged. Testids and mock-service contracts carry between them 1:1 — that is the acceptance mechanism.

Architecture

```
angular/src/app
├── core
│   ├── models          api.models.ts (frozen wire contracts) ·
│   session.models.ts (client state)
│   ├── api             SearchApi · TranslateApi · EntitiesApi ·
│   SummarizeApi · OcrApi
│   ├── stores          TriageStore · SessionStore · GoldCopyStore
│   (signals, in-memory)
│   ├── directives      PanelResize (shared drag-to-resize for both side
│   panels)
│   └── features
│       ├── triage       TriagePanel (queue + search) · SnippetHighlight
│       ├── viewer       ChatViewer · MessageBubble · OcrDialog
│       └── gold-copy     GoldCopyPanel · ProvenanceLine
```

For the system as a whole — services, certificates, identity, deployment topology, and the values only the target environment can supply — see [architecture.md](#). That document is written to be sufficient on its own; this one assumes you are in the code.

THREE THINGS IN HERE THAT ARE EASY TO UNDO BY ACCIDENT

The message stream is virtualized (`ChatViewer`, `cdk-virtual-scroll-viewport` from `@angular/cdk/scrolling`). The fixed-size strategy is deliberate: the autosize strategy lives in `@angular/cdk-experimental`, and adding an experimental package

that the enclave's npm mirror has to carry and the vulnerability gate has to clear is a bad trade for a deliverable the Government maintains. `itemSizePx` is measured, not guessed.

Two consequences worth knowing before touching that component. Anything that finds a message by `document.querySelector` will fail most of the time, because the target is usually not in the DOM — jump-to-evidence asks the viewport for the index first. And `tests/ng/virtual-scroll.spec.js` asserts a *bounded* DOM on a synthetic 2,000-message thread; a change that quietly reverts windowing fails there rather than in the enclave.

Render order is decided in the viewer, not taken from the service. Collection feeds do not guarantee chronological delivery — the audio-cut fixtures did not — so `orderedMessages` sorts, and the promoted transcript sorts chronologically regardless of which way the linguist is reading.

Nothing is persisted. No `localStorage`, `sessionStorage`, `IndexedDB` or cookies anywhere in either build. Panel widths, sort direction, viewer geometry and all officer work live in signals and reset on reload. This is a requirement (TDD Task 3), not an oversight; `SessionStore` is the seam if production needs otherwise.

- **Angular 21 (LTS), pinned exact** in `angular/package.json`; standalone components, signals, built-in control flow. No APIs newer than v17 — the pin is a stated assumption in the kickoff correspondence and an adjacent-major `ng update` is deliberately cheap.

- **Stores** hold all client state, in-memory only (the evaluated demo's performance claim — no persistence):
 - `TriageStore` — selected thread, focus-message (with a retrigger counter so repeat hits re-flash), and per-thread dispositions. Dispositions live here, not in the panel, because promoting a thread to gold marks it worked from the viewer side.
 - `SessionStore` — cached machine translations and entities (enrichment is thread-level, run once from the workflow strip), review verdicts, analyst notes, doc notes, and doc tags with the folded `tag → {threads, docs}` index that feeds the officer-tag facet.
 - `GoldCopyStore` — promoted thread-gold entries: the bench's single output (`hcd/one_output_model.md`). Snippet evidence clips were removed 1 Aug (logged deviation; single-commit revert if the Sponsor requires strict parity).
- **Enrichment is fixture-backed by default and gateway-backed by configuration.** `providers/model-gateway.js` speaks the OpenAI-compatible chat-completions wire format; supply `MODEL_ENDPOINT` and `MODEL_NAME` in the deployment `ConfigMap` and `MODEL_API_KEY` in the `Secret` (all three are environment-specific; no gateway address is committed), and translation and OCR answer from the enclave's live model instead of the seed fixtures — same contracts, no client rework. Translation sends full thread context; OCR uses vision on rasterized pages (`node scripts/rasterize-attachments.js`). Every failure path falls back to the fixture and logs. Provenance reports the engine (`<gateway-host>:<model>`), so the gold copy never misstates who translated what. Details: `deploy/README.md` §4.
- **API services** mirror the route contracts verbatim (`routes/*.js` headers are the source of truth; the contracts are frozen). Wire-format unit tests pin URL, params, and body shapes per service.
- **Escaping discipline:** corpus text renders as text nodes only — snippet highlighting builds `<mark>` spans from split text, never `innerHTML`.

Dev topology

```
npm install && npm --prefix angular install
npm run dev:ng          # mock services :5177 + ng serve :4200
(proxy.conf.json)
npm run dev             # the React reference app (:5273), same mock
services
```

Tests

```
npm --prefix angular run lint && npm --prefix angular run build && npm --
prefix angular test
npm run test:ng         # e2e vs the Angular build (Playwright boots both
servers)
npm test                # the original suite vs the React reference app
```

- `tests/*.spec.js` — the 7-spec Phase 2 acceptance suite (+ shared shell/services specs). It drives the browser by `testid`, so it runs against either app: the ported flows live verbatim inside `tests/ng/*.spec.js` and all 7 pass against the Angular build (see `project/evidence/README.md` for the coverage table).
- `tests/ng/` — Angular-build specs: the verbatim acceptance flows plus amendment coverage (queue progress, stack-clear, thread gold end-to-end). The React config ignores this directory.
- CI (`.github/workflows/angular-ci.yml`): lint → prod build → unit → e2e, plus the `package-validate` job (below) on every push.

Ground rules for new work

1. **data-testid values are the contract.** Never rename one; additions are fine (the amendment added `thread-*` / `queue-progress` / `stack-clear` testids — additive only).
2. **Mock contracts are frozen.** UI change? fine. Route shape change? that's a Sponsor conversation, not a refactor.

3. **Visual parity with the evaluated demo** — palette and layout tokens live in `styles.scss` as CSS custom properties (entity colors are part of the design contract). Parity baseline: `project/parity/`.

Deployment

The full packaging story (six images, kustomize manifests, probes/limits, statelessness, registry push, production cutover) is `deploy/README.md`. Short version:

```
./deploy/build-images.sh 0.1.0
kubectl apply -k deploy/k8s
kubectl port-forward svc/imax-spa 8443:8443
```

CI proves the set on a kind cluster every push (image builds → trivy CRITICAL gate → deploy → rollout → smoke through the SPA proxy chain). The production cutover swaps the nginx upstreams for enrichment-service endpoints — contracts frozen, no client rework.

Repo map

Path	What
<code>angular/</code>	Delivery SPA (this guide's subject)
<code>src/</code> , <code>index.html</code> , <code>vite.config.mjs</code>	React reference implementation (frozen spec)
<code>server.js</code> , <code>routes/</code> , <code>data/</code> , <code>static/</code>	Mock enrichment services + seed corpus (frozen contracts)
<code>tests/</code>	Playwright acceptance + Angular-build suites
<code>deploy/</code>	Dockerfiles, manifests, deployment guide
<code>hcd/</code>	Deliverable ① — personas, empathy maps, wireframes, workflow model
<code>docs/</code>	Deliverable ④ — this guide, the user guide, and <code>architecture.md</code> (Article XII)
<code>project/</code>	Deliverable ⑤/⑥ — risk register, burndown, evidence trail